East Swindon Development Group

Eastern Villages Swindon (South Marston), Primary Street

LANDSCAPE, ECOLOGY AND ARBORICULTURAL MANAGEMENT PLAN (LEAMP)

February 2025

This report and associated plans contain confidential information relating to badgers and should not be made available to members of the public.

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1.0 INTRODUCTION

1.1 This Landscape, Ecology and Arboricultural Management Plan (LEAMP) has been produced by FPCR Environment and Design Ltd. on behalf of the East Swindon Development Group.

1.2 It provides a management plan for the development at South Marston, in the context of, and with reference to the wider Eastern Villages development site (planning reference S/OUT/13/1555, condition 11). The planning consent is for outline permission:

'for a sustainable urban extension to the east of Swindon and north of the A420 of up to 2,380 dwellings together with a mixed use local centres/areas (including A1 retail up to 1,500 sq metres, services (A2), restaurants, pubs and takeaways (A3, A4, A5), business uses (B1) up to 1,000 sq metres, residential); community uses (D1); sheltered and/or care accommodation (C2/C3); two primary schools; green infrastructure including formal (including playing fields) and informal open space, landscape, biodiversity and amenity space; play space (including NEAPS/LEAPS/MUGAS); changing and sports facilities (including D2) and ancillary retail uses; sustainable drainage systems; highway, cycle and pedestrian routes; car parking; infrastructure (including utilities); engineering works including ground remodelling; demolition, site reclamation and removal of structures; the formation of new accesses from the A420, Old Vicarage Lane and Thornhill Road. | Land North Of A420 Eastern Villages Swindon'

1.3 The LEAMP is required to protect ecological receptors on Site and discharge Planning Condition 11 of the outline consent which states

11. Landscape, Ecology and Arboricultural Management Plan (LEAMP)

Each reserved matters application shall be accompanied by a Landscape, Ecology and Arboricultural Management Plan (LEAMP). The LEAMP shall be informed by the FLEAMPs approved under Condition 10 above and updated Phase 2 surveys where the last Phase 2 surveys undertaken for the site are more than 2 years old. The LEAMP shall include:

a) Details of the retention of any species receptor sites identified by the Phase 2 surveys for the site;

b) Details of protection for water courses and aquatic habitats against physical and chemical disturbance or pollution;

c) Details of long term objectives, extent and type of new planting; and

d) Details of any new habitat created on site, management responsibilities and maintenance schedules for all landscaped and habitat areas (except privately owned domestic gardens), and the sequencing for provision.

No development on each sub-phase shall commence until the LEAMP for that sub-phase has been approved in writing by the Local Planning Authority. Thereafter development on each sub-phase shall be progressed in accordance with the approved LEAMP. No development, works or other activities to the identified receptor sites shall be progressed other than in accordance with the works for ecological enhancement and management as set out in the approved LEAMP. The approved management and maintenance schedules shall be adhered to at all times. The submitted LEAMP shall be in accord with the approved Open Space Strategy.

Reason: To protect wildlife and supporting habitats and to secure opportunities for biodiversity enhancement.

1.4 This LEAMP provides details for the management of the area and habitats covered by the South Marston Primary Street as shown in Figure 1. This document builds on the habitat management prescriptions detailed in the FLEAMP for the South Marston Area[1] and references this throughout where relevant.

Site Context

1.5 The Site runs through the centre of the South Marston application boundary, from Old Vicarage Lane on the east to Thornhill Road in the west. The Site measures 13.44 ha.

1.6 Proposals are for the development of a Primary Street, which will be the main spine road for the wider South Marston development and all the development parcels which are part of it, with associated landscaping and access as shown in FPCR Landscape drawing (Drawing ref 1796-L-102-118 Landscape Design RevT).

2.0 ECOLOGICAL RECEPTORS

2.1 The Site has been subject to ongoing ecological surveys dating back to 2002. The most recent protected species surveys (badgers, bats, birds, dormice and reptiles) were conducted by FPCR in 2022 covering the South Marston area. The results of these surveys are detailed in separate reports, and have been used to inform the Site assessment and proposals.

2.2 The South Marston Site was subject to a pre-commencement Site walkover on the 28th February 2024, where changes in habitats and their potential to support protect species was assessed.

2.3 There were no significant changes to habitats and their suitability to support protected species, therefore the assessments carried out in the FLEAMP and protected species reports remain relevant.

2.4 Habitat lost as part of the development proposals, which offer varying levels of suitability to protected species, as discussed in the following sections, is shown in Figure 2.

Habitats

2.5 The area covered by the Site boundary comprises improved grassland fields bounded by hedgerows and associated seasonally wet and dry ditches, individual trees and areas of scrub and hardstanding.

Fauna

Great Crested Newt

2.6 The Site is of limited potential to GCN, restricted to potential hibernation opportunities within the hedgerow bases and associated ditches and areas of scrub. Foraging habitat is present within areas of less intensively managed grassland present along some of the hedgerow margins.

2.7 The likelihood of GCN using the low suitability habitats onsite is further reduced, as the majority of the Site is located outside of 250m of the nearest GCN pond.

1 Framework Landscape, Ecology and Arboricultural Management Plan (FLEAMP), Eastern Villages Swindon (South Marston), FPCR 2023

2.8 Impacts on GCN and compensation for habitat losses on-site and across the wider area will be dealt with through offsite enhancements secured by a Natural England District Level Licence. The licence (2024-00670-EPS-DLL) was issued by Natural England on the 11th June 2024. Following the receipt of the licence, all works within the Site can be undertaken without constraint. Although, it is recommended that removal of habitats suitable for GCN (e.g. hedgerows and associated ditches, scrub and trees) within 250m of GCN ponds should be supervised by an ecologist to prevent injury / killing to a protected species., in line with Natural England Best Practice Guidelines. These works should be carried out under suitable weather conditions, avoiding winter months, when GCN are likely hibernating in these locations.

2.9 Under the DLL scheme, no specific on-site enhancements are required, as these are delivered offsite. However, additional enhancements supplementing green infrastructure provision, as part of this application and the wider development, will result in additional enhancements for local amphibian populations.

Badger – Confidential (not to be released into the public domain)

2.10 The Site offers potential foraging habitats to badger *Meles meles*, comprising areas of improved grassland, scrub and hedgerows. Sett building habitats are restricted to hedgerows and associated ditches and areas of scrub within the Site.

2.11 A pre-commencement survey carried out by FPCR on the 28th February 2024, confirmed that the three disused setts previously identified within the Site boundary, were still disused.

2.12 Post development green infrastructure and enhancements, as part of this application and the wider development, will provide badger within the local area with new foraging and sett building opportunities.

2.13 As badgers are known to be active within the local area, precautionary working measures should be implemented during construction:

- A pre-commencement badger survey should be undertaken prior to Site clearance to ascertain whether any badger setts have been established in the interim.
- Any pipes greater than 250mm in diameter will be capped if they are left open overnight, thereby preventing badgers from becoming trapped;
- Any pits or trenches will be similarly covered overnight, or left with a suitable means of escape, e.g. wooden plank; failure to do so may result in entrapment of badgers and badgers creating a temporary sett within the trench, preventing works proceeding within 30m of this area.

2.14 All chemicals or materials that may harm badgers must be safely stored away from potential badger habitat and contained in such a way that they cannot be accessed or knocked over by badgers.

Birds

2.15 The hedgerow, scrub and tree habitats within the Site boundary provided nesting opportunities to a range of common and widespread species.

2.16 To avoid disturbance to breeding birds, where ground and structural vegetation is to be removed, it will be removed prior to the bird-breeding season (March to September). If this is not possible,

all ground and structural vegetation will be checked prior to removal by an experienced ecologist. If active nests are found, vegetation will be left untouched until all birds have fledged. Specific advice will be provided prior to undertaking the clearance. This would be a statutory requirement due to the protection of all nesting birds and their nests under the Wildlife and Countryside Act, 1981. A suitably qualified ecologist would supervise this.

2.17 If an active bird nest is found in vegetation or other feature/habitat to be removed, the area would be retained and suitably buffered. The buffer / exclusion zone is dependent on the bird species and location it is found in and will therefore be variable. Clearance operations would recommence only when the nest is no longer in use.

2.18 Post development green infrastructure and enhancements, as part of this application and the wider development, will provide birds within the local area with new foraging and nesting opportunities.

Bats

2.19 The update walkover survey carried out in February 2024 confirmed that there are 11 trees with bat potential (3 high potential trees, 5 moderate potential trees and 3 low potential trees) present within the Site Boundary. Some trees with bat potential are to be lost as part of the development proposals.

2.20 Trees within the Site boundary and wider South Marston Application Boundary were subjected to nocturnal surveys, relative to their bat roosting potential, between the 9th May 2022 and 1st August 2022. During these surveys no bat roosts were recorded within trees located in the Site boundary.

2.21 Trees with bat potential within the Site boundary were re-assessed regarding their bat roosting potential and subjected to aerial assessments in 2024, in line with the new bat survey guidelines. Where the trees are deemed unsafe to climb, they will be subjected to nocturnal surveys.

2.22 Current and previous tree surveys, tree bat potential and retention are summarised in Table 1 below.

Table 1: Bat Tree Potential and Survey Results Summary

Tree Ref	Bat Potential	Previous Survey Work & Results (2022)	Current Survey Work (2024)	Future Survey work (2024)	Retained (Y / N)
T2	Low	N/A – Low Potential	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	Yes
TG3a	Low	N/A – Low Potential	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No

Tree Ref	Bat Potential	Previous Survey Work & Results (2022)	Current Survey Work (2024)	Future Survey work (2024)	Retained (Y / N)
TG28a	High	3 Nocturnal surveys – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No
TG28b	Moderate	4 Nocturnal surveys – potential emergence on one survey, subjected to additional nocturnal, no roost detected	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	Yes
TG28c	Moderate	1 Nocturnal survey, further surveys cancelled as tree was set to be retained as part of proposals – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No
T34	Moderate	2 Nocturnal surveys – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No
T49	Moderate	2 Nocturnal surveys – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	Yes – TBC as part of Future Reserved Matters
T50	High	3 Nocturnal surveys – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	Yes – TBC as part of Future Reserved Matters
T54	High	3 Nocturnal surveys – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No

Tree Ref	Bat Potential	Previous Survey Work & Results (2022)	Current Survey Work (2024)	Future Survey work (2024)	Retained (Y / N)
T55	Moderate	2 Nocturnal surveys – no roost	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No
T56	Low	N/A – Low Potential	Reassessment & Climbing survey 31.07.24 identified features as PRF-I	PRF-I, pre-fell inspection required	No

2.23 During the climbing survey all trees within the Site boundary were assessed as PRF-I. Meaning that no additional climbing surveys were required. However, the trees will need to be subjected to a pre-fell inspection on the day of their removal. This will entail a climbing survey and sectional felling where appropriate.

2.24 In the event that a bat is found during the pre-fell inspection, works must stop and a Natural England EPSL will be required. Bespoke mitigation will be required.

2.25 To ensure retained suitable habitats continue to provide foraging and commuting opportunities to bats post-development, sensitive lighting will need to be adhered to during the construction phase. Measures to reduce impacts will include:

- The use of directional floodlighting to avoid spill.
- Avoidance of lighting potential dispersal corridors.
- The use of the lowest intensity possible.
- All luminaires should,
  a. Lack UV elements when manufactured. Metal halide, fluorescent sources would not be used.
  b. LED luminaires should be used where possible due to their sharp cut-off, lower intensity, good colour rendition and dimming capability.
  c. A warm white spectrum (ideally <2700Kelvin) should be adopted to reduce blue light component.

2.26 Six bat boxes will be installed prior to felling of trees with bat potential, to provide enhancements and compensation for the trees lost as part of the development.

2.27 The post development habitats and enhancements, as part of this application and the wider development, will provide new foraging and roosting opportunities for a range of bat species.

Riparian Mammals – Otter and Water Vole

2.28 The Site is unsuitable for water vole *Arvicola amphibius* and otter *Lutra lutra*, limited to seasonally wet heavily overshadowed ditches in association with hedgerow bases. These lack aquatic

vegetation, areas of permanent standing water and aquatic wildlife required by these species. Therefore, these species are not considered to pose a constraint to the development.

2.29 Post development green infrastructure and enhancements, as part of the wider development, will provide water vole and otter within the local area with new foraging and refuge opportunities.

Reptiles

2.30 The Site offers limited potential for reptiles, restricted to potential hibernation opportunities within the hedgerow bases and associated ditches and foraging habitat within the narrow less intensively managed hedgerow grassland margins. The remainder of the Site, comprises intensively managed short grassland habitats, lacking the structural variation required by reptiles.

2.31 The clearance works are anticipated to take place in Summer 2024, when hibernating reptiles are unlikely to be encountered within the hedgerow bases. Habitats clearance works will take place during suitable weather conditions (daytime temperature 9°C or higher), outside the dormant phase.

2.32 Post development green infrastructure and enhancements, as part of this application and the wider development, will provide reptiles within the local area with new foraging, basking and egg laying opportunities.

Hazel Dormouse

2.33 The Site was considered to be of limited value to dormice, restricted to the hedgerow, tree and scrub habitats within the Site.

2.34 Dormouse surveys in 2022 did not detect any evidence of occupation within the Site boundary. Therefore, this species is not considered to pose a constraint to the development.

2.35 Post development green infrastructure and enhancements, as part of this application and the wider development, will provide dormice with new foraging and nesting opportunities, should they become established in the local area in the future.

Summary

2.36 The Site supports potential receptor sites for the species mentioned above. Further survey work to ascertain the presence / likely absence of protected species is recommended and is ongoing. Precautionary working measures will be undertaken in line with licences and best practice guidelines to minimise the risk of impacting species during construction.

2.37 Post development green infrastructure will provide enhancements to a range of species.

3.0 WATER COURSES AND AQUATIC HABITATS PROTECTION MEASURES

3.1 The Site does not include any aquatic habitats or watercourses. In parts the Site falls within 15m of aquatic habitats. Site management protocols will be put in place to ensure that best practice measures are complied with. This will include methods to protect existing seasonally wet drains in association northern hedgerows throughout the Site.

3.2 Hedgerows and associated ditches, set to be retained under the development proposals, will be protected and buffered against accidental damage or encroachment through Heras panels placed at a 5m offset from the hedgerow and ditch.

3.3 The possibility of fuel spillages will be minimised through sound Site management.

3.4 The location of all works compounds will be agreed with the supervising ecologist. Any environmentally hazardous material will be kept in dedicated stores and storage tanks will have appropriate bunding.

3.5 To ensure construction works are undertaken in an environmentally responsible manner the best practice guidance listed below will be adhered to; these include Pollution Prevention Guidance (2012), some of which has been superseded by Guidance for Pollution Prevention (GPP):

- PPG 1: Understanding your environmental responsibilities - good environmental practices. A basic introduction to pollution prevention, with signposts to other PPGs and publications. (July 2013).
- GPP 2: Above ground oil storage tanks - For above ground oil storage, excluding oil refineries and distribution depots. (January 2018).
- PPG 3: Use and design of oil separators in surface water drainage systems - For identifying where an oil separator is required and, if so, what size and type of separator is appropriate. (April 2006).
- GPP 4: Treatment and disposal of wastewater where there is no connection to the public foul sewer - For selecting the correct sewage disposal, treatment and disposal options, and maintenance and legal requirements. Also, for what to have in mind, in terms of wastewater treatment, when buying a house. (November 2017).
- GPP 5: Works and maintenance in or near water - For construction or maintenance works near, in, or over water. (January 2017).
- PPG 6: Working at construction and demolition sites - For the construction and demolition industry. (2012).
- PPG 7: Safe storage - The safe operation of refuelling facilities for operators of liquid fuel refuelling facilities; it applies to all types of fixed refuelling facilities (July 2011).
- GPP 8: Safe storage and disposal of used oils - For storing and disposing of used oils. Applies to activities ranging from a single engine oil change to those of large industrial users. (July 2017).
- GPP 13 Vehicle washing and cleaning (April 2017) - For washing and cleaning any vehicle using automatic wash systems, high pressure or steam cleaners and washing by hand.
- PPG 18: Managing fire water and major spillages - For identifying equipment and techniques available to prevent damage to the water environment caused by fires and major spillages. (June 2000).
- GPP 21: Pollution incident response planning - For producing emergency pollution incident response plans to deal with accidents, spillages and fires. (July 2017).
- PPG 22: Incident response - dealing with spills or incident response - dealing with spills. (April 2011).
- PPG 23 Maintenance of structures - replaced by PPG 5, Works and maintenance in or near water.

- PPG 26 Safe storage - drums and intermediate bulk containers - For site operators of industrial and commercial premises storing and handling drums and intermediate bulk containers (IBCs) containing oil, chemicals or potentially polluting substances. (March 2011).

4.0 LANDSCAPE MEASURES

4.1 The landscape character and species composition of the different habitats as part of the wider site are detailed within the FLEAMP approved under condition 10.

4.2 The approved FLEAMP's aim was to inform the green infrastructure provision of the entire Site to ensure that the proposed development and the future management of the green infrastructure preserves and enhances the landscape, biodiversity and arboricultural resources in the long-term, in order to maximise its commitment to achieving Biodiversity Net Gain.

4.3 The following section details the long-term objectives, extent and type of new planting and the management responsibilities and maintenance schedules for all landscaped and habitat areas within the Primary Street Site boundary. The habitats proposed within this part of the Site form a part of the overall wider sites green infrastructure. Habitats types and management regimes are in line with the recommendations made within the FLEAMP.

4.4 Taylor Wimpey and Vistry will be responsible for future management and their appointed contractors will be responsible for the implementation of actions during construction period.

4.5 After construction, the Green Infrastructure will be handed over to a Management Company or other appropriate body.

Habitats within the Site

4.6 The Site's green infrastructure will comprise a combination of: retained hedgerows, trees and areas of improved grassland; newly planted native species rich hedgerows, native tree planting, native scrub planting, woodland planting, areas of amenity grassland, swales and SUDs comprising wetland wildflower grassland, pond edge planting and areas of meadow grassland.

4.7 The species composition and management principals are detailed in Tables 2 and 3 below.

Table 2: Detailed Framework Management Principles

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
1	**Retained Hedgerows** Maintain & enhance existing native species hedgerows	• Protect existing hedgerows during construction in accordance with the CEMP and BS 5837:2012, or as superseded.	Implement sympathetic management • Cut 50% of each length on a rotational basis between January and February (so that autumn fruits are available for wildlife) and again avoiding frost periods. • Cut to no less 2m high. • Retain a minimum 1.5m margin at base and cut every 3 years to promote dense cover. • Laying hedge once new plantings have established between November and the end of February to maintain a dense structure
2	**Retained Trees** Maintain existing native trees	• Protect existing trees during construction in accordance with the CEMP and BS 5837:2012, or as superseded.	Implement sympathetic management • Within publicly accessible areas, and especially within proximity of defined footpath routes within green space, trees would be checked annually, after major storms and at any other interval deemed necessary following inspection for essential remedial works in the interest of public safety, should be undertaken as advised by a suitably qualified Arboriculturist.
3	**Retained Improved grassland** Maintain existing areas of grassland	• Protect areas of grassland to be retained using Heras fencing	• Implement the same management as outlined in Item 8 below. • Seed any areas of damaged grassland using Emorsgate EL1 Flowering Lawn Mix as detained in Item 8 below.
4	**Hedgerows** New native species rich hedgerows	Create new hedgerows using species which reflect the species-composition of the existing hedges • Prepare vegetation-free planting strip. • Plant using staggered double row, 500mm between rows at a rate of 6 plants/m. • Plant 60-90cm bare root whips. • Planting mix to reflect species composition of existing hedges <u>Native hedgerow</u> *Acer campestre* Field Maple (20%) *Carpinus betulus* Hornbeam (5%) *Corylus avellana* Hazel (40%) *Ilex aquifolium* Holly (5%) *Ligustrum vulgare* Wild privet (30%) • Protect with shrub guards • Seed with a suitable hedgerow seed mix	Implement sympathetic management • Upon establishment (year 2-3) cut 50% of each length on a rotational basis between January and February (so that autumn fruits are available for wildlife) and again avoiding frost periods. • Cut to no less 2m high. • Retain a minimum 1.5m margin at base and cut every 3 years to promote dense cover. • Laying hedge once new plantings have established between November and the end of February to maintain a dense structure
5	**Native Tree planting** Individual tree planting throughout the green infrastructure to reflect the different character areas along the Primary Street	New tree planting to be in line with the Strategic Design Code: Tree Strategy. • Cut and clear any existing vegetation once prior to planting • Plant standard trees (as detailed in the landscaping plans) between October and March, avoiding periods of inundation or prolonged ground frost • Mulch a 1m diameter circle around the tree stem to a minimum depth of 75mm • Protect establishing tree planting with guards • Hand-weed to supress weed growth Tree species and number of trees to be planted will depend on the Character Area they falls within. Species will comprise:	• Within publicly accessible areas, and especially within proximity of defined footpath routes within green space, trees will be checked annually, after major storms and at any other interval deemed necessary following inspection for essential remedial works with works undertaken as advised by a suitably qualified Arboriculturist. • All works will be undertaken according to best practice as set out in BS 3998 Tree Work – Recommendations and using qualified tree surgeons/or arborists to satisfy a duty of care. • All vegetation and, particularly, woody vegetation proposed for clearance should be removed outside of the main bird-breeding season (March – September inclusive) as all birds are protected under the Wildlife and Countryside Act, 1981 (as amended) whilst on the nest. Where this is not possible, vegetation should be checked for the

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		Acer campestre Field maple (5) *Acer platanoides Columnulare* Norway maple (2) *Aesculus hippocastanum* Horse chestnut (2) *Alnus glutinosa* Black alder (6) *Betula ermanii* Erman's birch (4) *Betula pendula* Silver birch (15) *Betula pubescens* Downy birch (3) *Carpinus betulus* Hornbeam (7) *Castanea sativa* Sweet chestnut (4) *Corylus colurna* Turkish hazel (4) *Liquidambar styraciflua* Sweetgum (19) *Prunus avium* Wild cherry (1) *Quercus robur* English oak (9) *Salix alba* White willow (5) *Salix caprea* Goat willow (14) *Sorbus aucuparia* Rowan (7) *Tilia x europaea Pallida* Common lime (7)	presence of nesting birds by a suitably qualified Ecologist. • Arisings from tree works will, where practical, be used to create small dead wood piles within the woodland planting. Alternatively, where this is not possible, they can be chipped to provide mulch for use in amenity planting areas or removed from site. • Standing dead wood will be retained in situ where possible to provide habitat. Where it is not safe to retain this, the deadwood piles and recumbent logs will be created/ placed at marginal areas, providing habitat for a range of fauna including saproxylic invertebrates and fungi, plants, small mammals, common reptiles and amphibians.

6	**Scrub**
	Scrub will be allowed to grow to maturity and, in time, scrub to succeed to woodland where desirable
		Shrub planting should include native species. Species for inclusion within strategic green space to be chosen carefully to ensure they are appropriate for the area and will contribute to the overall aims of strategy to provide a net gain in biodiversity value. Typical species lists are outlined below:
		
		<u>Scrub:</u>
		Cornus sanguinea Common Dogwood (362)
		Corylus avellana Hazel (362)
		Euonymus europaeus European spindle (271)
		Ilex aquifolium Holly (362)
		Ligustrum vulgare Wild privet (271)
		Viburnum lantana Wayfaring tree (181)
		
		A suitable grassland mix, will be surface sown in autumn or early spring onto a prepared seed bed.
		
		To support the establishment and long-term success of new planting the following measures should be adopted:
		• Shrubs are to be checked annually when they are in full leaf to ensure that they are thriving. Defects requiring remedial works will be recorded.
		• Shrubs which die or do not establish will be replaced the first planting season following each annual inspection during the first five years after planting
		• Cut areas of scrub in a rotation, aiming to retain all ages. Scrub typically matures in
			• Early promotion of canopy species including weed management and understorey scrub management.
			• Tree shelters and stakes to be inspected twice annually as part of the general maintenance visits and adjusted accordingly. Damaged ties or stakes must be replaced. When the trees are established and can support themselves the ties should be carefully removed and the stakes removed/cut down to ground level. This operation is likely to be required after three to five years dependant on establishment rates, stability and growing conditions.
			• Periodic monitoring and control/removal of non-native species as necessary, including any Schedule 9 species.
			• Replacements of like-for-like in event of failure or promotion of neighbouring established specimens as deemed necessary.
			
			• All tree guards, tubes, ties and stakes that have not biodegraded naturally are to be removed from site once plants are mature enough to survive browsing.
			• Periodic trimming/management of understorey scrub in planting groups including:
			  ○ Regular cutting to create dense stands with graded margin; and
			  ○ Infrequent management of selective stands to promote more open stands (including thinning of stands and light trimming to aid succession of canopy species).
			  ○ Plantings protected with rabbit guards as necessary.
			  ○ Monitoring and installation of fencing or other protection as deemed necessary. Periodic monitoring and control of browsing pressure.
			
			In every management year:

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		15 years, so cut 1/5 every third year. • Newly shrubs to be strimmed in a sensitive manner, to avoid damage to any habitats to be retained, from around the base of the plants until the point it is determined that they are tall enough to tolerate weed competition. If the shrubs show signs of poor growth or reduced vigour an application of the appropriate fertiliser should be carried out. If shrubs do not respond to a treatment of fertiliser, further inspections are required, including the ground conditions for signs of compaction, contamination, and poor-quality topsoil. Any problems uncovered with the soil must be remediated. Should remediation works not resolve the problem a replacement shrub may need to be planted. • During establishment shrubs should be watered regularly during prolonged dry periods. • All vegetation removal should be timed outside the bird nesting season, i.e. undertaken between September and February. Should vegetation removal be required outside this timeframe an ecologist should be consulted prior to works commencing. • Shrubs are vulnerable to browsing by livestock, deer and rabbits during establishment. Protect with spiral guards or tubes (using non-plastic and biodegradable where possible) and are removed when the plants are mature enough to survive browsing.	• Retention of trimmings following management (where not diseased) in brash piles and windrows. • Retention of standing deadwood including branches and whole trees (where not diseased).

7	**Woodland and Woodland Edge Planting**
	New woodland will be allowed to grow to maturity and in time woodland edge planting will succeed into woodland where desirable
		Woodland planting should include native species. Species for inclusion within strategic green space to be chosen carefully to ensure they are appropriate for the area and will contribute to the overall aims of strategy to provide a net gain in biodiversity value. Typical species lists are outlined below:
		
		<u>Woodland Planting:</u>
		Acer campestre Field maple (195)
		Alnus glutinosa Black alder (34)
		Betula pendula Silver birch (34)
		Betula pubescens Downy birch (23)
		Cornus sanguinea Common Dogwood (57)
		Corylus avellana Hazel (114)
		Crataegus monogyna Common hawthorn (172)
		Euonymus europaeus European spindle (57)
		Ilex aquifolium Holly (57)
		Pinus sylvestris Scots pine (80)
		Prunus avium Wild cherry (23)
		Prunus spinosa Blackthorn (23)
		Quercus robur English oak (148)
		Sorbus aucuparia Rowan (34)
		Tila cordata Small-leaved lime (125)
			Early promotion of canopy species including weed management and understorey management.
			• Tree shelters and stakes to be inspected twice annually as part of the general maintenance visits and adjusted accordingly. Damaged ties or stakes must be replaced. When the trees are established and can support themselves the ties should be carefully removed and the stakes removed/cut down to ground level. This operation is likely to be required after three to five years dependant on establishment rates, stability and growing conditions.
			• Periodic monitoring and control/removal of non-native species as necessary, including any Schedule 9 species.
			• Replacements of like-for-like in event of failure or promotion of neighbouring established specimens as deemed necessary.
			
			• All tree guards, tubes, ties and stakes that have not biodegraded naturally are to be removed from site once plants are mature enough to survive browsing.
			• Periodic trimming/management of understorey scrub in planting groups including:
			  ○ Regular cutting to create dense stands with graded margin; and
			  ○ Infrequent management of selective stands to promote more open stands (including thinning of stands and light trimming to aid succession of canopy species).
			  ○ Plantings protected with rabbit guards as necessary.
			  ○ Monitoring and installation of fencing or other protection as deemed necessary. Periodic monitoring and control of browsing pressure.
			
			In every management year:
			• Retention of trimmings following management (where not diseased) in brash piles and windrows.
			• Retention of standing deadwood including branches and whole trees (where not diseased).

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		Woodland Edge Planting: *Acer campestre* Field maple (131) *Cornus sanguinea* Common Dogwood (174) *Corylus avellana* Hazel (131) *Crataegus monogyna* Common hawthorn (219) *Euonymus europaeus* European spindle (44) *Ilex aquifolium* Holly (44) *Malus sylvestris* Crab apple (44) *Rosa canina* Dog rose (87) A suitable woodland mix, such as the Emorsgate EW1 Woodland mix or similar agreed, will be surface sown in autumn or early spring onto a prepared seed bed. • Trees are to be checked annually when they are in full leaf to ensure that they are thriving. Defects requiring remedial works will be recorded. • Trees which die or do not establish will be replaced the first planting season following each annual inspection during the first five years after planting • Newly planted trees to be strimmed in a sensitive manner, to avoid damage to any habitats to be retained, from around the base of the plants until the point it is determined that they are tall enough to tolerate weed competition. If the trees show signs of poor growth or reduced vigour an application of the appropriate fertiliser should be carried out. If trees do not respond to a treatment of fertiliser, further inspections are required, including the ground conditions for signs of compaction, contamination, and poor-quality topsoil. Any problems uncovered with the soil must be remediated. Should remediation works not resolve the problem a replacement tree may need to be planted. • During establishment trees should be watered regularly during prolonged dry periods. • All vegetation removal should be timed outside the bird nesting season, i.e. undertaken between September and February. Should vegetation removal be required outside this timeframe an ecologist should be consulted prior to works commencing. • Trees are vulnerable to browsing by livestock, deer and rabbits during establishment. Protect with spiral guards or tubes (using non-plastic and biodegradable where possible) and are removed when the plants are mature enough to survive browsing.	
8	**Amenity Grassland –** amenity grass verges adjacent to the Primary Street road	• Areas to be sown will be first rotovated and raked or harrowed to produce a medium fine, firm tilth. Areas will be planted with amenity grass mix EL1 Flowering Lawn Mix at a rate of 4g/m2, in the autumn or spring, selecting a time when the soil is moist and can be worked. Wildflowers: *Betonica officinalis* Betony (1.0%) *Centaurea nigra* Common Knapweed (2.4%) *Galium verum* Lady's Bedstraw (1.2%) *Leontodon hispidus* Rough Hawkbit (0.8%) *Lotus corniculatus* Common Bird's-foot-Trefoil (2.4%)	Year 1 & 2 • Regular cutting would seek to ensure a sward height of between 40-50mm during the growing season to ensure resilience to drought and avoid the need for chemical fertilizers and herbicides. Thereafter • Cut between March and October to achieve desired height.

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
		Plantago lanceolata Ribwort Plantain (2.8%) *Primula veris* Cowslip (0.8%) *Ranunculus acris* Meadow Buttercup (1.2%) *Silaum silaus* Pepper Saxifrage (0.4%) *Vicia cracca* Tufted Vetch (0.4%) *Medicago lupulina* Black Medic (Ag) (5.0%) *Trifolium repens* Small Leaved White Clover (Ag) (5.0%) Grasses: *Agrostis capillaris* Common Bent (Ag) (8% *Carex flacca* Glaucous Sedge (1%) *Cynosurus cristatus* Crested Dog's-tail (Ag) (39%) *Festuca rubra* Red Fescue (Ag) (28%) *Phleum bertolonii* Smaller Cat's-tail (4%)	
9	**Swale-** create areas of wet wildflower grassland within swales along the Primary Street road and within SUDs, beneficial to wildlife	• Area along road verge and SUDs to be sown with Emorsgate EP1F Wildflowers for Wetlands see mix at a rate of 1.5g/m2 in the autumn or spring, selecting a time when the soil is moist and can be worked. • Strim to height of 75mm once in late August and again in mid-October Wildflowers: *Angelica sylvestris* Wild Angelica (5.00%) *Centurea nigra* Common Knapweed (13.00%) *Cruciata laevipes* Crosswort (5.00%) *Dipsacus fullonum* Wild teasel (5.00%) *Eupatorium cannabinum* Hemp Agrimony (2.00%) *Filipendula ulmaria* Meadowsweet (6.00%) *Geum rivale* Water Avens (3.00%) *Iris pseudacorus* Yellow Iris (20.00%) *Lathyrus pratensis* Meadow Vetchling (4.00%) *Lotus pedunculatus* Greater Birdsfoot Trefoil (4.00%) *Lycopus europaeus* Gypsywort (1.00%) *Oenanthe pimpinelloides* Corky-fruited Water-dropwort (1.00%) *Plantago lanceolata* Ribwort Plantain (6.00%) *Prunella vulgaris* Selfheal (1.00%) *Ranunculus acris* Meadow Buttercup (5.00%) *Silene dioica* Red Campion (14.00%) *Silene flos-cuculi* Ragged Robin (5.00%)	Year 1 & 2 • Post sowing (first growing season) – only mow flush of annual weeds by mowing in mid to late summer • Removing the clippings • Mow through to the end of March of the following year • Dig out any residual perennial weeds such as docks Thereafter • Strim to height of 75mm once in late August and again in mid-October • In August, leave cuttings for 3-5 days to allow to dry, for re-dispersal of seeds. • Do not apply any fertiliser
10	**Pond Edge Planting**	• Small areas of pond edge planting will be created in the SUDs in the east of the Site.	• Cut back and remove short sections of vegetation every 2-3 years in rotation,

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
	Areas of pond edge planting will be created within the large SUDs on the east of the Site of benefit to wildlife	These areas will be seeded with Emorsgate EP1 mix at a rate of 4g/m^2 in the autumn or spring, selecting a time when the soil is moist and can be worked. Wildflowers: *Angelica sylvestris* — Wild angelica (1.00%) *Carex divulsa ssp divulsa* — Grey sedge (1.00%) *Centurea nigra* — Common knapweed (2.40%) *Cruciata laevipes* — Crosswort (0.80%) *Dipsacus fullonum* — Wild teasel (0.60%) *Eupatorium cannabinum* — Hemp agrimony (0.10%) *Filipendula ulmaria* — Meadowsweet (1.70%) *Galium album* — Hedge bedstraw (1.00%) *Geum rivale* — Water avens (0.60%) *Geranium pyrenaicum* — Hedgerow crane's-bill (0.20%) *Iris pseudacorus* — Yellow iris (4.00%) *Lathyrus pratensis* — Meadow vetchling (0.80%) *Lythrum salicaria* — Purple loosestrife (0.50%) *Lycopus europaeus* — Gypsywort (0.20%) *Oenanthe pimpinelloides* — Corky-fruited water-dropwort (0.20%) *Plantago lanceolata* — Ribwort plantain (0.60%) *Prunella vulgaris* — Selfheal (0.20%) *Ranunculus acris* — Meadow buttercup (1.00%) *Silene dioica* — Red campion (2.80%) *Silene flos-cuculi* — Ragged robin (0.70%) Grasses: *Agrostis capillaris* — Common Bent (w) (4.00%) *Anthoxanthum odoratum* — Sweet vernal-grass (w) (4.00%) *Carex divulsa subsp. divulsa* — Grey sedge (w) (1.60%) *Cynosurus cristatus* — Crested dogstail (34.40%) *Deschampsia cespitosa* — Tufted hair-grass (w) (1.60%) *Festuca rubra* — Red fescue (20.00%) *Hordeum secalinum* — Meadow barley (w) (4.00%)s *Poa trivialis* — Rough-stalked meadow-grass (8.00%) *Schedonorus arundinaceus* — Tall fescue (2.40%)	between September and November. • Within ponds and detention basins, vegetation should be removed as a wedge. • Dense stands of single species may benefit from selective thinning.
11	**Meadow Grassland** Create and maintain a species rich grassland	• Remove existing vegetation and prepare fine, consolidated seedbed. • Seed areas with Emorsgate EM5 Meadow Mixture at a rate of 4g/m2 mid-September to end of 1st week in October.	Implement traditional management Year 2 after sowing

ITEM	FRAMEWORK MANAGEMENT OBJECTIVE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – CONSTRUCTION PHASE	FRAMEWORK MANAGEMENT TO ACHIEVE OBJECTIVE – POST CONSTRUCTION
	of benefit to wildlife	Wildflowers: *Achillea Millefolium* Yarrow (0.6%) *Agrimonia eupatoria* Agrimony (0.8%) *Betonica officinalis* Betony (0.4%) *Centaurea nigra* Common knapweed (3%) *Daucus carota* Wild carrot (0.6%) *Galium verum* Lady's bedstraw (1.6%) *Knautia arvensis* Field scabious (1.6%) *Leontodon hispidus* Rough hawkbit (0.1%) *Leucanthemum vulgare* Oxeye daisy (2%) *Lotus corniculatus* Birdsfoot trefoil (0.2%) *Malva moschata* Musk mallow (2.4%) *Plantago lanceolata* Ribwort plantain (2.6%) *Poterium sanguisorba* Salad burnet (1.2%) *Primula versis* Cowslip (0.8%) *Ranunculus acris* Meadow buttercup (0.2%) *Ranunculus bulbosus* Bulbous buttercup (0.1%) *Rumex acetosa* Common sorrel (1%) *Silene vulgaris* Bladder campion (0.4%) *Vicia cracca* Tufted vetch (0.4%) Grasses: *Anthoxanthum odoratum* Sweet vernal-grass (w) (1.6%) *Briza media* Quaking grass (w) (3.2%) *Cynosurus cristatus* Crested dogstail (56%) *Festuca rubra* Red fescue (10.4%) *Agrostis capillaris* Common bent (8%) *Trisetum flavescens* Yellow oat-grass (w) (0.8%) • Post sowing (first growing season) – control flush of annual weeds by mowing regularly and removing the clippings. • Mow when sward height reaches 100mm; do not mow lower than 50mm.	• From June to mid-September leave un-mown to flower and set seed freely. • From mid-September to end of May mow regularly and remove clippings. • Arisings to be relocated, in part, to designated areas on the southern boundary • Do not apply any fertiliser. • Spot treat any undesirable ruderal species such as docks and thistles using a contact herbicide applied via a hand-held weed wipe or applied with a hand brush – do not spot treat with a knapsack sprayer.

Table 3: Framework Post construction and Management Programme

Habitat/ Feature	Regular Operations/Works (Years 1-5)	Anticipated Frequency of Visits/Operations (see note below)												Frequency, notes
		J	F	M	A	M	J	J	A	S	O	N	D	
1. Hedgerow (existing)	Cutting													As required
2. Trees (existing)	Cutting													As required
	Maintenance													Annually to year 5
3. Grassland (existing)	Long-term management has been implemented													As required
4. Hedgerow (new)	Planting													Year 1 only
	Seeding with hedgerow mix													Year 1 only
	Cutting													From year 3 and as necessary subject to monitoring
5. Native Tree Planting	Planting													Year 1 only
	The recommended planting species have been used													Year 1 only
	The guards are in place													Years 1-4
	Plants have been replaced if less than 75% establishment of original planting													Years 2-5
6. Scrub	Planting													Year one only
	Maintenance													Annually to year 5
	Selective thinning & trimming a sinuous edge													From year 2 – to promote regeneration of seedlings and saplings
	Control of pernicious weeds/undesirable ground cover													Annually as needed to Year 5 then thereafter as necessary monitoring
7. Woodland	Planting													Year 1 only
	Seeding with woodland grassland mix													Year 1 only
	Maintenance													Annually to year 5
	Selective thinning & trimming a sinuous edge													Year 5 onward (thinning), 1/3 to ½ on rotation Every 3-5 years.
	Control of pernicious weeds/undesirable ground cover													As necessary subject to monitoring.
8. Amenity Grassland	Areas of amenity grassland have been sown													Year 1 only
	The recommended seed mix has been used													Year 1 only
	The first-year management has been implemented													Year 1 only

Habitat/ Feature	Regular Operations/Works (Years 1-5)	Anticipated Frequency of Visits/Operations (see note below)												Frequency, notes
		J	F	M	A	M	J	J	A	S	O	N	D	
	Long-term management has been implemented													As required
9. Swale / SUDs	Areas of wetland meadow grassland have been sown													Year 1
	Cutting													In late summer/ autumn
	Scrub/weed control													Year 1 as required for weed control (including in spring), otherwise annually from year 2
10. Pond Edge Planting	Areas of wetland meadow grassland have been sown													Year 1
	Cutting													In late summer/ autumn
11. Meadow Grassland	Areas of meadow grassland have been sown													Year 1
	Cutting													In late summer/ autumn

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Key

Primary Street Site Boundary

Wider Development Site Boundary

client
East Swindon Development Group

project
Primary Street,
Eastern Villages Swindon (South Marston)

drawing title
SITE LOCATION PLAN

scale @ A3
1:10000

drawn
APR / PH

issue date
18/2/2025

drawing / figure number
Figure 1

rev
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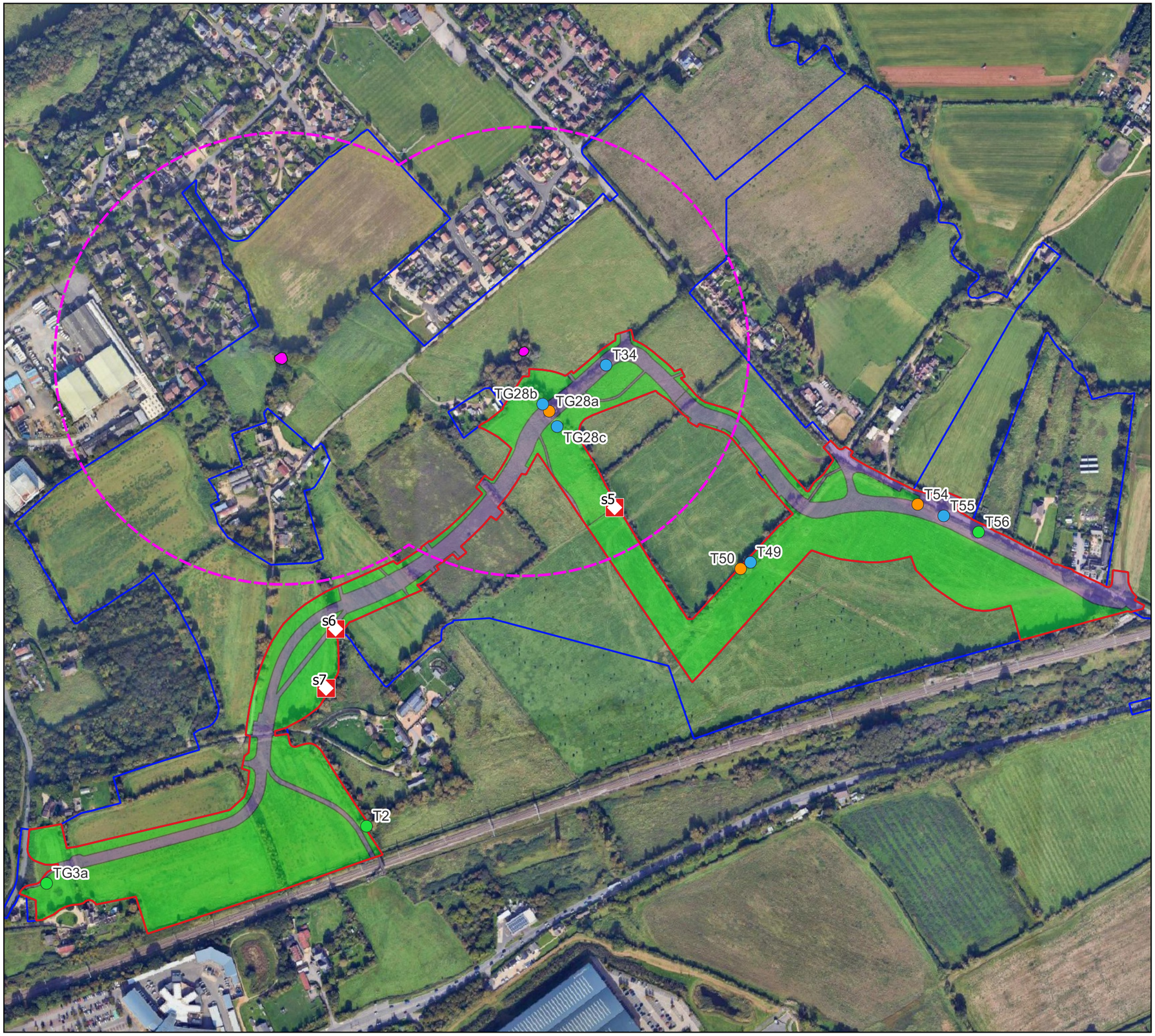

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Key

Primary Street Site Boundary

Wider Development Site Boundary

Habitats

Habitats Lost

Habitats Retained / Enhanced

Bats

Trees with Bat Potential within Primary Street Site Boundary

High

Moderate

Low

GCN

GCN Waterbodies Near Primary Street

250m Breeding Pond Buffer within Primary Street Site Boundary

Badger

Badger Setts within Primary Street Site Boundary

Disused Sett

30m Disused Badger Sett Buffer within Primary Street Boundary

The habitat loss / retention & enhancements shown on this plan are indicative and are based on the Landscape Plans. Given the scale of this plan, it should be considered in combination with the Tree Retention Plan and Landscape Plans for this Site - on which this plan is based.

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client
East Swindon Development Group

project
Primary Street,
Eastern Villages Swindon (South Marston)

drawing title
HABITAT / SPECIES RECEPTOR SITE IMPACT PLAN

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1:4500

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Figure 2

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